Pressure Effects on Free Flame Velocity

MAE 5430: Technical Memorandum

March 27th, 2026

Diya Mehta

Executive Summary

In this technical memorandum, the effect of pressure on laminar free flame is explored for methane and ethylene, representing alkane and alkene fuels respectively. Flame speed depends on thermal diffusivity and reaction rates, both of which are pressure-dependent, with flame speed decreasing with pressure following a power law relationship for both fuels.

Model

Cantera's free flame solver was used to model a one-dimensional, adiabatic, premixed flame, isolating pressure effects across two fuels. Key assumptions and their effects are summarized below:

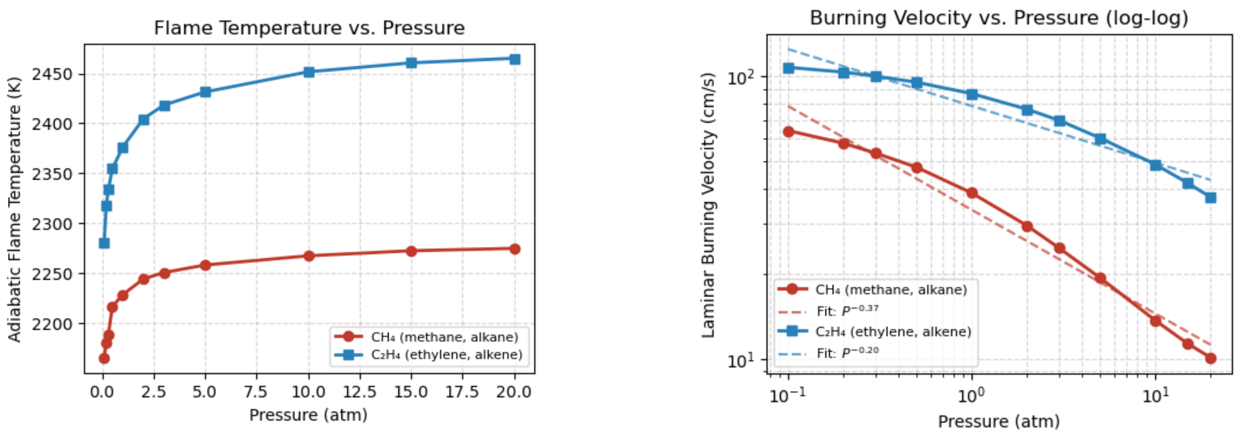
Assumption	**Effect**
Laminar flow	Does not account for turbulent flow or associated energy loss
Premixed and Stoichiometric	Assumes perfect mixture, no excess fuel or oxidizer, and assumes all species react
Free propagation	Assumes no heat loss to boundaries; real confined flames experience wall heat losses that reduce flame speed and can cause quenching
Ideal gas	Gas is not compressible and enthalpy would be different with a non-ideal gas
One-dimensional	Cannot capture multi-dimensional phenomena such as flame curvature, stretch, or cellular instabilities that are present in real flames
Steady-state	Assumes constant flame properties, real flames can be unsteady and the model does not reflect that.
Mixture-averaged transport	Approximates species diffusion via weighted binary coefficients; may underpredict diffusion of fast light species
Chemical mechanisms	Limited to 53 species and 325 reactions; excludes interactions with species outside the GRI-Mech 3.0 set

Table 1: Model Assumptions and Their Effects

This model, though having many assumptions, still provides a good approximation of flame velocity. The reactions are determined based on what is kinetically and thermodynamically possible within the chosen chemical mechanism. GRI-Mech 3.0 was selected as it is specifically validated for natural gas combustion, containing 53 species and 325 reactions, making it well-suited for methane and ethylene flames within the pressure range studied.

One major modeling choice was setting the equivalence ratio to stoichiometric. This is a reasonable assumption because many rockets and non-air-breathing engines aim to operate at the precise fuel-to-oxidizer ratio required for complete combustion, minimizing the formation of minor species and unburned reactants. Stoichiometric conditions yield the maximum adiabatic flame temperature, providing an upper bound for flame temperature and flame speed, which is useful for understanding the theoretical limits of combustion performance. Additionally, this choice simplifies calculations by reducing chemical complexity and allows for clearer interpretation of pressure-dependent kinetic effects without the added variable of the mixture. These results are particularly relevant to rocket propulsion, as flame speed directly affects how quickly and completely combustion occurs in the chamber, which in turn influences the chamber pressure that drives thrust.

Data



(a) Adiabatic Flame Temperature

(b) Flame Speed

Figure 1: Free Flame Characteristics vs. Pressure

The overall trend of flame speed is that it decreases as pressure increases, following an approximate power law relationship. As pressure increases, the higher collision frequency leads to a slight increase in adiabatic flame temperature due to suppressed dissociation. Intuitively, one might expect flame speed to increase with pressure, especially given the corresponding rise in temperature. However, at elevated pressures, terminating reactions become increasingly dominant over chain-branching reactions, consuming reactive radicals and shifting the radical balance away from propagation. This kinetic effect overwhelms any temperature-driven increase, resulting in a net decrease in laminar flame speed.

When comparing different fuels at the same pressure, a higher adiabatic flame temperature generally corresponds to a higher flame speed, which explains why the alkene fuel exhibits a higher flame speed than the alkane fuel. In this case, the governing factor is fuel chemistry and its influence on reaction rates, whereas in the pressure-dependent trend, the dominant mechanism is the shift toward terminating reactions at high pressure.

Limitations

The most significant limitation of this analysis is the restricted pressure range over which the model is validated. Cantera's freely available flame models and underlying chemical mechanisms are primarily validated against experimental data within the 1–10 atm range; for pressures below this range, the model's accuracy diminishes due to limited validation data, while for pressures above 10 atm, the reliability of reaction rates, transport properties, and thermodynamic data becomes increasingly uncertain. Consequently, predictions outside the 1–10 atm range may exhibit inconsistencies or unreliable trends. Additionally, the one-dimensional assumption introduces further limitations, as it cannot account for viscous effects, turbulence, or multi-dimensional flow phenomena that are ubiquitous in real combustion systems. Turbulent flames, for example, can exhibit flame speeds orders of magnitude higher than laminar flames due to enhanced mixing and increased flame surface area, effects entirely absent from this 1D laminar framework. Together, these constraints limit the model's applicability for predicting flame behavior in practical combustion devices such as gas turbines, internal combustion engines, and rocket propulsion systems, where pressures often exceed 10 atm and turbulent flow dominates.